

Vector-Space Optimization Framework Based on Projected Contraction Condition for Control Design with Input Saturation

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- Existing Approaches
- Common Limitations
- Summary and Motivation for Proposed Method

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Contraction Theory is a powerful tool for analyzing the stability of nonlinear systems by examining **differential dynamics**, providing insights into the behavior of **trajectories** (not points) and their convergence properties [1].

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) \quad \rightarrow \quad \text{diff. dyn.} \quad \frac{d}{dt}\delta\mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\delta\mathbf{x}. \quad (1)$$

Typically used Lyapunov function is defined with respect to $\delta\mathbf{x}$ rather than $\mathbf{x}$, as follows:

$$V = \mathbf{x}^\top \mathbf{P} \mathbf{x} \quad \rightarrow \quad V = \delta\mathbf{x}^\top \mathbf{M}(\mathbf{x}, t)\delta\mathbf{x}. \quad (2)$$

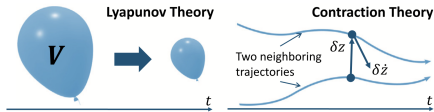


Figure: Lyapunov theory versus contraction theory [2].

Therefore, in control design, **Contraction Theory** provides:

- **Exponential Convergence:** Trajectories globally exponentially converge to each other regardless of initial conditions.
- **Control Design:** Owing to nature of differential dynamics, **LPV control design** methods can be applied.
- **and more...**

Existing Approaches Control Contraction Metric (CCM)

Control Contraction Metric (CCM) provides a framework for designing stabilizing controllers [3].

Consider a control-affine system and Lyapunov function candidate as follows:

$$\frac{d}{dt} \mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \mathbf{u} \rightarrow \text{diff. dyn. } \frac{d}{dt} \delta \mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \mathbf{B}}{\partial \mathbf{x}} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}, \text{ and } V = \delta \mathbf{x}^\top \mathbf{M}(\mathbf{x}, t) \delta \mathbf{x}, \quad (3)$$

Theorem (Control Contraction Metric (CCM) [3])

For a given system, if there exists a bounded metric $\mathbf{M}(\mathbf{x}, t)$ such that holds

$$\frac{d}{dt} \mathbf{M} + 2 \text{sym} \left(\mathbf{M} \frac{\partial \mathbf{f}}{\partial \mathbf{x}} + \frac{\partial \mathbf{B}}{\partial \mathbf{x}} \mathbf{M} \right) \leq -2\alpha \mathbf{M}, \quad \left(\frac{d}{dt} V \leq -2\alpha V \right), \quad (4)$$

for all $\delta \mathbf{x} \neq 0$ such that $\delta \mathbf{x}^\top \mathbf{M} \mathbf{B} = 0$, then the system is **universally exponentially stabilizable** via state feedback.

By using $\mathbf{M}$, a feedback controller $\mathbf{u}$ can be established considering the closest point on the **geodesic** between the current state and the target state in the **Riemannian space** defined by $\mathbf{M}$, i.e., $\delta \mathbf{u} = k(\mathbf{x}, \delta \mathbf{x}, \mathbf{u}, t)$.

How to find $\mathbf{M}$?

For interested operating domain, CCM $\mathbf{M}$ should be solved in advance, via LMIs or Sum-of-Squares (SOS) programming.

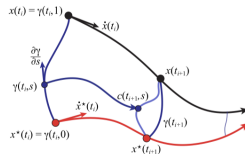


Figure: Control Contraction Metric (CCM) [3].

ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) is an alternative approach to CCM, which calculates the contraction metric $\mathbf{M}$ in step-wise manner [4].

By defining $\delta \mathbf{u} = k \rightarrow \mathbf{u} = K(\mathbf{x} - \mathbf{x}_d)$, $K = \mathbf{Y}\mathbf{M}$, the contraction metric can be obtained by solving **linear matrix inequalities (LMIs)** with decision variable $\mathbf{W}^{-1} := \mathbf{M}$ [4].

According to Theorem 2.4 in [2], by **minimizing the condition number** χ of $\mathbf{M}$, the **steady state error** can be reduced.

$$\downarrow \chi(\text{condition number of } \mathbf{M}) \rightarrow \downarrow \text{steady state error} \quad (5)$$

The control input can be simply obtained by

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \quad (\mathbf{M} \text{ is obtained by solving LMIs which minimizes } \chi.) \quad (6)$$

Main features of CV-STEM are as follows:

- Easy to implement (point-wise convex condition).
- Due to the use of LMIs, it still should be solved in advance.
- According to the objective function, typically, $\mathbf{M}$ is designed to be identity matrix and ν is maximized, which results in **a simple state feedback controller with a large proportional gain.**

What if the control input is saturated?

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t)\text{sat}(\mathbf{u}) \quad (7)$$

Typically, contraction theory-based control design methods does not consider input constraints, which can lead to **undesirable behavior** in practical applications.

Summary of the discussion on contraction theory-based control design methods:

- CCM is a powerful tool providing the existence of stabilizing controllers.
- CV-STEM provides a more computationally efficient approach to find contraction metric M in a step-wise manner.
- However, both CCM and CV-STEM still ...
 - Computationally expensive (should be solved in advance).
 - Difficult to handle constraints, such as input saturation.

The proposed approach is ...

- Solve the contraction problem in an online manner by projecting the contraction condition into a vector space, resulting in a **point-wise convex condition in vector space**.
- Extend the method to the systems under **input constraints**.

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Problem Formulation

Consider a control-affine system under input saturation and external disturbance:

$$\frac{d}{dt} \mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \text{sat } \mathbf{u} + \mathbf{d}(t), \quad (8)$$

assuming that

- The system functions are smooth implying Lipschitz continuity.
- The control effectiveness matrix $\mathbf{B}(\mathbf{x}, t)$ is full column rank.
- The disturbance is bounded as $\|\mathbf{d}(t)\| \leq \bar{d}$.

The desired trajectory is supposed to be given as $\mathbf{x}_d(t)$ and $\mathbf{u}_d(t)$, which satisfies the system dynamics:

$$\frac{d}{dt} \mathbf{x}_d = \mathbf{f}(\mathbf{x}_d, t) + \mathbf{B}(\mathbf{x}_d, t) \mathbf{u}_d, \quad (9)$$

without input saturation and external disturbance.

Problem Statement

The objective is to design a stabilizing controller $\mathbf{u}$ such that the error dynamics $\mathbf{e} = \mathbf{x} - \mathbf{x}_d$ is contracting under the input saturation and external disturbance.

Revisit of Existing Approach (CV-STEM)

Using state-dependent coefficient (SDC) formulation, the error dynamics can be derived in LPV form as follows:

$$\begin{aligned} \frac{d}{dt} \mathbf{e} &= \underbrace{\mathbf{f} - \mathbf{f}_d + (\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d}_{=:\mathbb{A}(\mathbf{x}, \mathbf{x}_d)\mathbf{e}} + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \underbrace{\mathbf{B} \text{sat } \mathbf{u} - \mathbf{u}}_{=:\phi(\mathbf{u})} + \mathbf{d} \\ &= \mathbb{A}(\mathbf{x}, \mathbf{x}_d)\mathbf{e} + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}, \end{aligned} \quad (10)$$

Consider the following control law:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \quad (11)$$

where $\mathbf{M}$ is a **contraction metric** which will be obtained.

If the control law can be realized without input saturation, the error dynamics can be rewritten as follows:

$$\frac{d}{dt} \mathbf{e} = \left(\mathbb{A}(\mathbf{x}, \mathbf{x}_d) - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \right) \mathbf{e} + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}. \quad (12)$$

Then, the steady tracking error is defined as follows:

$$\lim_{t \rightarrow \infty} \|\mathbf{e}(t)\| \leq \frac{\bar{d}}{\alpha} \sqrt{\chi}, \quad (\text{Theorem 2.4 in [2] and } \chi \text{ is cond. number of } \mathbf{M}) \quad (13)$$

Revisit of Existing Approach (CV-STEM)

The LMI problem can be formulated to minimize the condition number χ of the contraction metric $\mathbf{M}$ as follows [4]:

$$\begin{aligned} & \min_{\overline{\mathbf{W}}, \chi, \nu} \chi + \lambda \nu \\ \text{s.t. } & \begin{cases} -\frac{d}{dt} \overline{\mathbf{W}} + 2 \text{sym}(\mathbb{A} \overline{\mathbf{W}}) - 2\nu \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \preceq -2\alpha \overline{\mathbf{W}}, \\ \mathbf{I}_n \preceq \overline{\mathbf{W}} \preceq \chi \mathbf{I}_n, \end{cases} \end{aligned} \quad (14)$$

where $\mathbf{M} =: \overline{\mathbf{W}}^{-1}$, $\overline{\mathbf{W}} = \nu \mathbf{W}$ is normalized $\mathbf{W}$, and ν is decision variable for magnitude of the contraction metric, and λ is penalty parameter for ν .

However, the existing approach requires

- To solve the LMI problem to obtain the contraction metric for the interested operating domain in advance.

Our Approach

We relax the contraction condition by projecting it onto the tracking error space.

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The contraction condition from Theorem 2.4 [2] is represented as follows:

$$\frac{d}{dt} \mathbf{M} + 2 \text{sym}(\mathbf{M}\mathbb{A}) - 2\mathbf{M}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \preceq -2\alpha \mathbf{M}. \quad (15)$$

We note that eventually the metric is used as a feedback gain matrix as

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}\mathbf{e}. \quad (16)$$

Therefore, the metric is only used in the subspace spanned by the error $\mathbf{e} = \mathbf{x} - \mathbf{x}_d$.

Thus, we can project the contraction condition onto $\mathbf{e}$, which results in a point-wise convex condition with low computational burden.

$$\mathbf{y}^\top \left(-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \right) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, \quad (17)$$

where $\mathbf{y} := \mathbf{M}\mathbf{e}$ denotes a projected contraction metric onto the $\mathbf{e}$, serving as a new optimization variable.

Objective Function (Condition Number): According to Theorem 2.4 in [2], the minimization of the condition number χ of $\mathbf{M}$ can be achieved by minimizing the following objective function:

$$J = \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e}. \quad (18)$$

Contraction Condition: Rewrite the projected contraction condition as

$$\mathbf{y}^\top \left(-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \right) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbf{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0. \quad (19)$$

Energy Condition: Minimizing the condition number may result in a very small control input. To avoid this issue, we add the energy condition as follows:

$$\left(\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e} \right)^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, \quad (20)$$

where $\mu \in \mathbb{R}_{>0}$ denotes the desired energy level, and $\epsilon \in \mathbb{R}_{>0}$ and $s \in \mathbb{R}_{\geq 0}$ denote the allowable small energy variation and a slack variable.

Additional Constraints: Any constraints can be added to the optimization problem.

Resulting Optimization Problem:

$$\begin{aligned} & \min_{\mathbf{y}, s} \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e} + \lambda s \\ \text{s.t. } & \begin{cases} \mathbf{y}^\top (-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, \\ (\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e})^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, \end{cases} \end{aligned} \quad (21)$$

λ to achieve sufficient control effort and μ to determine the desired input energy

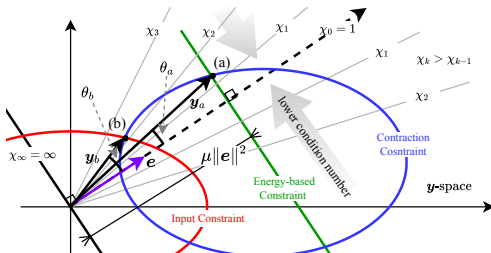


Figure: Example illustration of the proposed method with eclipse-shaped input saturation constraint.

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We employed the Lorenz system to validate the proposed method, as follows:

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=: \mathbf{x}} = \underbrace{\begin{pmatrix} \sigma(y - x) \\ x(\rho - z) - y \\ xy - \beta z \end{pmatrix}}_{=: \mathbf{f}(\mathbf{x})} + \text{sat} \left(\underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=: \mathbf{u}} \right) + \mathbf{d}(t), \quad (22)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$.

The saturation function and external disturbance are defined as follows:

$$\text{sat}(\mathbf{u}) := \begin{cases} \mathbf{u}, & \text{if } \|\mathbf{u}\| \leq \bar{u}, \\ \bar{u} \frac{\mathbf{u}}{\|\mathbf{u}\|}, & \text{otherwise,} \end{cases}, \quad \text{and} \quad \mathbf{d}(t) := 1000(H(t - 0.1) - H(t - 0.11)) \cdot \mathbf{1}_{3 \times 1}, \quad (23)$$

where $\bar{u} = 160$ and $H(\cdot)$ is the Heaviside step function.

The desired trajectory is given by simulating the system with desired control law defined as follows:

$$\mathbf{u}_d(t) := -\mathbf{f}(\mathbf{x}_d(t)) - 2(\mathbf{x}_d(t) - \mathbf{r}), \quad \text{where } \mathbf{r} = \begin{pmatrix} -8.485 \\ 8.485 \\ 27 \end{pmatrix}. \quad (24)$$

Simulation Results

Table: Controllers for comparative study.

(C ₁) [—]	Proposed controller.
(C ₂) [—]	CV-STEM ([5]).
(C ₃) [—]	Desired controller for nominal model, i.e., $\mathbf{u} = \mathbf{u}_d$.

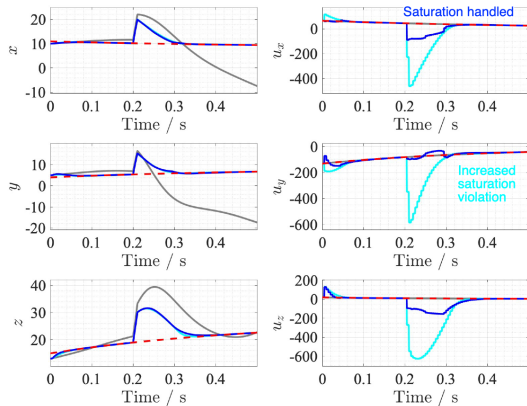


Figure: Trajectories tracking results and control inputs

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IFAC

- Currently, effective-space formulation and input constraints are considered.
- The input constraint will be removed and the effective-space formulation will be more rigorously justified.

Future work

- Additional constraints, such as state constraints and CBF-motivated constraints, and input constraints will be considered.
- Real-time implementation of the proposed method will be investigated.

Towards Deep Learning

- Using obtained $\mathbf{y}$, we can extract the contraction metric $\mathbf{M}$.
- Using neural networks, we can learn the mapping from the state $\mathbf{x}$ to the contraction metric $\mathbf{M}$.
- Hence, if we can justify that the obtained metric is a valid contraction metric under imposed constraints, we can use the learned metric for control design and analysis.

Thank you for your attention!

- [1] W. LOHMILLER and J.-J. E. SLOITINE, "On contraction analysis for non-linear systems," *Automatica*, vol. 34, no. 6, pp. 683–696, 1998. [Online]. Available: [https://doi.org/10.1016/S0005-1098\(98\)00019-3](https://doi.org/10.1016/S0005-1098(98)00019-3)
- [2] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," *Annual Reviews in Control*, vol. 52, pp. 135–169, 2021. [Online]. Available: <https://doi.org/10.1016/j.arcontrol.2021.10.001>
- [3] I. R. Manchester and J.-J. E. Slotine, "Control contraction metrics: Convex and intrinsic criteria for nonlinear feedback design," *IEEE Transactions on Automatic Control*, vol. 62, no. 6, pp. 3046–3053, 2017. [Online]. Available: <https://doi.org/10.1109/TAC.2017.2668380>
- [4] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Neural stochastic contraction metrics for learning-based control and estimation," *IEEE Control Systems Letters*, vol. 5, no. 5, pp. 1825–1830, 2021. [Online]. Available: <https://doi.org/10.1109/LCSYS.2020.3046529>
- [5] H. Tsukamoto and S.-J. Chung, "Neural contraction metrics for robust estimation and control: A convex optimization approach," *IEEE Control Systems Letters*, vol. 5, no. 1, pp. 211–216, 2021. [Online]. Available: <https://doi.org/10.1109/LCSYS.2020.3001646>